RGO1 – RENORMALIZED CONSTRUCTION AND NON-GAUSSIANITY IN QED3

Abstract. This stage outline organizes the model estimates needed for the common tuned QED3 construction and its positive fourth cumulant. One linked map connects three existing focused outlines and six planned treatments. The precise theorem contracts remain in QEO1 and the existing focused works. Continuum convergence and component theorems are explicit external inputs.

Contents

1. Introduction and target statement 1
Proof overview 3
2. Focused works and their contracts 4
3. MBO1. Maxwell block analysis 5
4. FBO1. Chamber fermion locality and integration 6
5. RNO1. Extraction, reconstruction and normalized contraction 7
6. S15. The exact nonlinear renormalization step 8
7. T01. Agreement across covers and one tuned pair 9
8. T09. Terminal estimates and the iterated measure 10
9. T11. Two-sided partitions and common pressure 11
10. T04. Free coefficient bounds and continuum comparison 12
11. B7. A compact probe with positive fourth trace 13
12. T05. Buffered source placement 14
13. T06. Exact zero-flux and harmonic identities 15
14. T07. The exact source-shift and Wick identity 16
15. Q4R. Normalized reference response 17
16. Q1. Finite all-flux inputs for volume transfer 18
17. T17. Uniform products on arbitrary construction volumes 19
18. A1. Continuum parents, recovery and moment limits 20
19. B2. Gaussian domination in the same components 21
20. B5. Distinct products control conditional expectations 22
21. T16. One common choice of constants and charge range 23

1. Introduction and target statement

RGO1 is the renormalization and response stage beneath the overall QEO1 plan. It coordinates nine focused scopes. The full tensor OS assembly remains in QEO1. All model estimates retain the fixed chamber cutoff, norms, domains and conditional parameter inequalities of their linked contracts.

Overall outline QEO1. The package directory follows the large map. External statement links use the accompanying QEO1 PDF in this same output folder.

A3. Positive fourth cumulants for the same construction

MODEL-SPECIFIC. Overview Target.

Target of this stage. Fix the standard-weight chamber cutoff, $0 < \rho < 1$, and $0 < \bar{m} \le 1$, as in QEO1/D0. The construction selects one real bare mass and vacuum pair $P_N(e) = (m_{0,N}(e), \epsilon^{\mathrm{v}}_{0,N}(e))$, independently of the outer construction volume. Its total fine mass is $\bar{m}L^{-N} + m_{0,N}(e)$. Every interacting construction, reference and transfer law uses this pair. The free comparisons retain the unshifted positive reference mass.

For a real compact smooth two-form β, set

$$f = dd^*\beta, \qquad b(f) = -6\,\mathrm{Tr}[(\bar{m} + \gamma\cdot\partial)^{-1} i\gamma\cdot d^*\beta]^4.$$

The probe theorem B7 supplies a choice with $b(f) > 0$. The desired conclusion is that there are $C_f < \infty$, $\sigma > 0$, and $e_f > 0$, independent of the chosen parent and its component, such that for $0 < e \le e_f$, every volume-first parent law μ of this construction has

$$|\kappa_4^{\mu_\xi}(f) - e^4 b(f)| \le C_f e^{4+\sigma}$$

on one full-measure set of its translation-ergodic components. Reducing e_f so $C_f e^\sigma \le b(f)/2$ gives $\kappa_4^{\mu_\xi}(f) \ge e^4 b(f)/2 > 0$. Here $\kappa_4(X) = \mathbb{E}[(X - \mathbb{E}X)^4] - 3\mathbb{E}[(X - \mathbb{E}X)^2]^2$.

The main estimate. The workhorse is T17: a bound for every finite set of distinct translated Wick probes, uniform in the construction volume and fine mesh. A parent fourth-moment estimate alone does not give the component conclusion.

Inputs and scope. Use A1 to retain each prescribed parent and its moment limits, B2 for the Gaussian bound on the same components, and B5 to pass translated-product bounds to conditional expectations. The final parameter choice is T16. The tensor OS conclusions remain in QEO1's structural branch. Intersect its component set with the one obtained here.

Exact contracts. QEO1/A3, QEO1/T18, and QEO1/Z. This is the target of a working proof outline, not a new proof. The diagram's ordinary statement pages describe the assignments; their external links open the complete existing contracts.

RGO1 – Renormalized construction and non-Gaussianity in QED3

Definitions/setup · Model-specific · Interface (general)

QEO1 > RGO1 > nine focused scopes. Standing inputs: the fixed chamber cutoff and each contract's norms, domains and parameter inequalities. Existing theorem identifiers are retained. Boxes link to ordinary pages, then to the full contracts.

Model-specific
MBO1. Maxwell block estimates
Uniform local means, fluctuation coordinates and one-step Gaussian integration.
Statement p. 5
Proof estimate: 13–18 pp.

Model-specific
RNO1. Extraction and contraction
Bounded reconstruction and normalized scaling in the declared activity space.
Statement p. 7
Proof estimate: 19–25 pp.

Model-specific
FBO1. Fermion kernels and integration
Local chamber inverses, weighted completion and localized determinant gains.
Statement p. 6
Proof estimate: 23–31 pp.

all inputs

Model-specific
S15. Exact nonlinear step
External-slot density and a convergent activity map with size and Lipschitz bounds.
Statement p. 8
Proof estimate: 26–34 pp. with S12–S14

Model-specific
T01. Covers and common tuning
Exact cover agreement transports one selected bare pair to the reference tori.
Statement p. 9
Proof estimate: 21–28 pp. with S16–S18

Model-specific
T06. Zero-flux identity
Exact-curvature Gaussian and harmonic-twist normalization; retained sector projection.
Statement p. 15
Proof estimate: Included in RRO1 identities

all inputs

Model-specific
T09. Terminal estimates and measure
Terminal analytic bounds; sector domination; the iterated Borel density and complement.
Statement p. 10
Proof estimate: 16–22 pp. with T08

Model-specific
T05. Place the reflected sources
At the selected reference scales, buffered sources fit in cells and have orthogonal reflected curvatures.
Statement p. 14
Proof estimate: 11–15 pp. with T06–T07

all inputs

Model-specific
T11. Partitions and pressure
Two-sided partition bounds and a common two-volume matter defect for that pair.
Statement p. 11
Proof estimate: 9–13 pp. with T10

Model-specific
T07. Exact Wick source identity
Apply the finite source shift to the actual model probes, with their normalization.
Statement p. 16
Proof estimate: Included in RRO1 identities

Model-specific
T04. Free coefficient comparison
Uniform free coefficients, continuum fourth-trace comparison and second-order cancellation.
Statement p. 12
Proof estimate: 16–22 pp. with B7

all inputs

Model-specific
Q1. Finite all-flux inputs
Finite positivity, Gaussian domination, chessboards and trace comparison for the volume transfer.
Statement p. 18
Proof estimate: Imported fixed-spacing work

Model-specific
Q4R. Reference response
Mixed normalized derivatives: one-cell, multi-cell and complement estimates for the same pair.
Statement p. 17
Proof estimate: 16–22 pp.

all inputs

Model-specific
T17. Uniform translated products
Main estimate: uniform bounds for every finite set of distinct translated probes on arbitrary construction volumes.
Statement p. 19
Proof estimate: 9–13 pp.

Interface
B7. Positive compact probe
Choose a compact probe with a strictly positive continuum fourth coefficient.
Statement p. 13
Proof estimate: Included in PRO1

Model-specific
A1. Continuum parent and moments
Realize each prescribed parent and pass finite products along its joint cutoff sequence.
Statement p. 20
Proof estimate: Imported CEM1; T18 checks

Model-specific
T16. Admit one common scheme
Realize the conditional inequalities on one positive charge interval; no feedback to their proofs.
Statement p. 23
Proof estimate: Shared parameter verification

Interface
B2. The same component hypotheses
Retain the Gaussian constant on almost every component of the same parent.
Statement p. 21
Proof estimate: Imported ECM1

Interface
B5. Conditional Wick moments
A general distinct-product criterion controls invariant conditional expectations.
Statement p. 22
Proof estimate: 4–7 pp. with T18/A3

all inputs

Model-specific
A3. Positive component cumulants
For every prescribed parent, the same components have a positive fourth cumulant.
Statement p. 2
Proof estimate: Final application in VTO1

Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.
Click a box for its statement. Return links are navigation, not proof dependencies.

This is a working proof map. Conditional estimates are proved before T16 verifies one common admissible scheme. T01 supplies the same pair throughout the terminal, pressure and response arguments. Early chamber and free-response work can proceed independently; later reconstructed-connection clauses retain their photon inputs. Source identities T05–T07 belong to RRO1, but T06 is used before its response checkpoint Q4R. Budgets overlap and are not additive. T04 imports only the free kernel part of FBO1. T05 uses the selected reference geometry; its finite geometric proof may be made conditionally. The photon determinant defect and flux tail are proved inside T17. Independent imported results are placed near their uses. Some direct uses of earlier conditional inequalities by T16 are omitted from the overview and retained on its statement page and in the complete edge inventory.

2. Focused works and their contracts

The diagram separates mathematical checkpoints where their uses differ. The nine rows below are document scopes. Several checkpoints can belong to one focused work, and membership does not give a dependency arrow.

2.1. **Block integration and the common construction.**

MBO1: Maxwell block analysis. Existing focused outline; the proof companion is MBM1. Its output is MBO1.
FBO1: chamber fermion analysis. Existing focused outline. Its kernel, integration and determinant tasks are FBO1.
RNO1: relevant extraction and contraction. Existing focused outline. Its separate reconstruction and contraction outputs are RNO1.
NLO1: the exact nonlinear step. Planned; not created. The existing contracts are QEO1/S12–S15, linked from S15. The density and nonlinear convergence remain distinct proof tasks within this work.
CVO1: cover agreement and common tuning. Planned; not created. The existing contracts are QEO1/S16–S18 and T01, linked from T01.
TPO1: terminal measure and pressure. Planned; not created. The existing contracts are QEO1/T08–T11, linked from T09 and T11. The finite identity T06 is imported from the early part of RRO1.

2.2. **Response and non-Gaussianity.**

PRO1: a positive probe and free coefficients. Planned; not created. The existing contracts are QEO1/B7 and T02–T04, linked from B7 and T04. These calculations can precede the interacting trajectory.
RRO1: exact sources and reference response. Planned; not created. The early source, sector and Wick contracts are QEO1/T05–T07. The response contract is Q4R, with proof tasks T12–T15. Follow T05–T07 and Q4R. The early identity T06 is used in TPO1; the later response uses the terminal outputs. These are directed uses of different results and do not create a circular proof.
VTO1: arbitrary-volume transfer and component cumulants. Planned; not created. The existing contracts are QEO1/T17–T18, B5 and A3, linked from T17, B5 and A3. The finite all-flux estimates Q1, continuum input A1 and component theorem B2 remain external imports.

2.3. **Use with the overall plan.**

QEO1 retains its five groups and twenty-one finer proof assignments. RGO1 collects the model work in those assignments into nine independently maintainable focused scopes. It supplies neither a replacement OS theorem nor an additional proof of the imported structural results.

Keep QEO1's PDF beside this PDF for the external statement links. Each diagram box opens its local description first; that page identifies the complete contract, source proof span, inputs and next use. The six planned outlines have no separate files or asserted proof status.

Page estimates are inherited from the QEO1 assignments and their QEM1 source spans. They count complete exposition, not remaining work. Shared setup and imported results occur in more than one budget, so the numbers must not be summed.

3. MBO1. Maxwell block analysis

Model-specific. Overview Target.

Output. Construct the constrained Abelian block minimizer and fluctuation coordinates, prove their differentiated locality, and obtain the exact one-step Gaussian energy decomposition. Retain the local-gauge estimates, weakening bounds and actual domain margins. Exact operator identities on covers concern the unweakened operators.

Inputs and use. The cochain averages, gauge constraints and mesh normalization are fixed in MBO1/MB0. The domain conclusions have additional charge inequalities. The exported estimates enter local fermion integration and the exact density in S15. The iterated photon measure is the later task T09.

Focused outline and source. MBO1 is an existing focused outline under RGO1. Its principal contract is MBT; the actual Gaussian split is S04. The full Maxwell proof is maintained as MBM1. See also QEO1/S03– S05 and M21; QEM1 Section 10, pp. 89–102. The complete proof estimate is 13–18 pages.

4. FBO1. Chamber fermion locality and integration

Model-specific. Overview Target.

Output. Prove the light/heavy spectral separation and analytic local inverse estimates, including their covariant Hölder and weakening bounds. Establish weighted Berezin completion and the localized determinant logarithm with its small-field gain. Keep the first-level chamber measure and free determinant normalizations.

Inputs and use. The kernel work starts from the chamber operator and averaging maps. Its reconstructed-connection application also imports the specified photon estimates from MBO1. This dependence does not prevent the free chamber analysis from starting independently. The resulting integration and determinant estimates enter S15.

Focused outline and source. FBO1 is an existing focused outline. Its target FT distinguishes the required character conventions; uniform unitary-character kernel estimates do not automatically extend the fixed-character determinant conclusions. Full contracts: QEO1/S01– S02, S05–S08. QEM1 Sections 9 and 11, pp. 77–89 and 102–113. The complete proof estimate is 23–31 pages.

5. RNO1. Extraction, reconstruction and normalized contraction

Model-specific. Overview Target.

Output. In one declared coefficient norm and analytic domain, define the relevant scalar extraction and prove bounded reconstruction. For the normalized remainder, establish the small-polymer gauge and complex-domain estimates that give contraction under scaling and reblocking.

Inputs and use. The chamber stencil, field domains, symmetry assumptions and weighted algebra estimates are explicit inputs. Bounded reconstruction and normalized contraction are separate outputs: the former is not a prerequisite for the latter. Both are used in the complete nonlinear map S15.

Focused outline and source. RNO1 is an existing focused outline. See its setup D, reconstruction S10, gauge argument G and contraction S11. Full master contracts: QEO1/S09– S11, with M20. QEM1 Section 12, pp. 113–133. The complete proof estimate is 19–25 pages, with shared setup counted once.

6. S15. The exact nonlinear renormalization step

Model-specific. Overview Target.

Required result. Construct the actual new activity by finite localization and a convergent connected logarithm. Prove its exact density identity, symmetry, size bounds and coupling-difference bounds. The external connection remains an independent variable. Retain the distinction between the external-slot identity and the identity restricted to zero external connection.

Proof assignment. First establish the normalized one-step density S12, using the Maxwell coordinates and fermion integration. Then prove localization S13, connected convergence S14, and the complete map S15. The inputs include the activity norm and domains of RNO1, its reconstruction and contraction estimates, and the correct initial chamber weight. The Lipschitz conclusion is needed for tuning.

Allocation. NLO1 is planned; its separate outline has not been created. These four contracts remain in QEO1. QEM1 Sections 13–14, pp. 133–159: allow 9–12 pages for the density and 17–22 for nonlinear closure, with common definitions counted once.

7. T01. Agreement across covers and one tuned pair

Model-specific. Overview Target.

Required results. Construct compatible projected primitives and supported rows, then prove equality of the actual local steps and extracted scalars under the specified odd-cover lifts. Use that equality to transport the pair selected by the mixed-boundary fixed point on the top torus to the smaller reference tori.

Inputs and conclusion. The projected primitive construction can begin before the nonlinear map. Equality of the complete steps requires S15. The fixed point evolves activities forward and masses backward; it selects a particular pair $P_N(e)$, independent of twists and sources. Its mass bound does not characterize an arbitrary admissible replacement pair. Use precisely this pair in T09, T11, Q4R, and T17.

Allocation and contracts. CVO1 is planned; its separate outline has not been created. Full statements: QEO1/S16– S18 and T01. QEM1 Sections 15–16, pp. 159–178. Allow 14–18 pages for exact cover agreement and 7–10 for selection and transfer of the common pair.

8. T09. Terminal estimates and the iterated measure

Model-specific. Overview Target.

Required results. For the trajectory selected in T01, prove the terminal inverse neighborhoods and insertion bounds in T08. Then identify the terminal density for every bounded Borel coarse observable, including the iterated photon coordinates, Jacobian, energy isometry and curvature-row counts.

The delicate input. The zero-flux Gaussian estimate retains its sector projection. Its proof uses T06; conditioning the all-flux Gaussian bound is not a replacement. The downward change of variables must be iterated; the one-step Maxwell identity alone does not identify this measure. Retain the positive reference mass separately from the real mass interaction.

Uses and allocation. These outputs enter both pressure T11 and reference response Q4R. Marked insertion bounds in T08 have an additional response use. TPO1 is planned and groups these tasks with pressure; it does not introduce a pressure input to the response proof. Full contract: QEO1/T09. QEM1 Sections 21.1–22.3, approximately pp. 205–219. Allow 8–11 pages for terminal operators and 8–11 for sector domination and the iterated measure.

9. T11. Two-sided partitions and common pressure

Model-specific. Overview Target.

Required result. Prove the free-pressure and twist estimates, control the terminal remainder and obtain positive upper and lower partition bounds. Cancel the common bulk term on the two reference tori using exactly the pair selected by T01. The resulting matter defect is uniform in the fine mesh and admitted charge.

Inputs and use. Use the terminal measure and complement bounds of T09 and the free calculation T10. The latter can begin before the interacting trajectory is constructed. Pressure is an input to the all-volume transfer T17; it does not replace the normalized response theorem Q4R.

Allocation and contracts. This completes planned TPO1. QEO1/T11 gives the precise estimate; Q3 states the combined common construction and pressure output. QEM1 Sections 22.4–22.6, pp. 219–227. Allow 9–13 pages with T10.

10. T04. Free coefficient bounds and continuum comparison

Model-specific. Overview Target.

Required result. Compare the free chamber fourth coefficient with its continuum Schatten-four trace, uniformly at the specified mesh and reference scales. Prove the separate two-photon cancellation needed for the second coefficient. For the resulting coefficients $c_{2,\eta}, c_{4,\eta}$, retain both uniform bounds and the comparison of $c_{4,\eta} - 3c_{2,\eta}^2$ with $b(f)$.

Independence and use. This calculation imports the free chamber symbol and kernel estimates S01 from FBO1; it assumes no interacting trajectory. Its comparison enters T17 and the component-cumulant conclusion. The smaller coefficient bounds needed inside Q4R come from its own one-cell analyticity estimate T12. There is no T04-to-Q4R prerequisite.

Allocation and contracts. PRO1 is planned and groups this calculation with B7. Full statements: QEO1/T02– T04. QEM1 Sections 17–18, pp. 178–194. Allow 16–22 pages for the whole positive/free assignment.

11. B7. A compact probe with positive fourth trace

Interface. Overview Target.

Required result. For the continuum massive Dirac inverse, construct a real compact smooth two-form β such that $f = dd^*\beta$ has the strictly positive coefficient $b(f)$ displayed in A3. Verify the trace class required by that expression and preserve the compact-support construction.
Use. The sign enters the final positive-cumulant conclusion A3. The normalized product estimate T17 can be stated for a probe without this sign assumption. The positive probe construction can be undertaken independently of tuning and interacting response.
Allocation and source. This is the continuum part of planned PRO1, with T04. Its full hypotheses and proof assignment are in QEO1/B7; QEM1 Section 17. The 16–22-page PRO1 budget includes this argument.

12. T05. Buffered source placement

Model-specific. Overview Target.

Required result. Construct the centered discrete bond probe from a fixed compact smooth β, identify its physical curvature pairing and variance, and place its buffered support in an open reference cell. Reflections into distinct cells must give orthogonal source curvatures.

Inputs and use. Specify the mesh, reference side, support margin and reflected cell geometry. The application uses the selected reference and stopping scales of T01; the geometric calculation itself can be made conditionally at prescribed scales. This finite geometric construction is distinct from the terminal marked insertion bounds in T08. It supplies the actual probes for T07 and Q4R.

Allocation and source. This is the early source checkpoint in planned RRO1. Full contract: QEO1/T05. QEM1 Section 19. The source/sector/Wick assignment T05–T07 has a combined 11–15-page budget.

13. T06. Exact zero-flux and harmonic identities

Model-specific. Overview Target.

Required result. Unfold the zero-flux model into its exact-curvature Gaussian and harmonic-twist integral with all determinant and volume normalizations retained. Identify the sector projection used in the Gaussian estimate.

Uses. This finite identity supplies an input to T09 and to the exact source calculation T07. It can be proved before common tuning. Its placement in the source-and-response work does not make it depend on the later response theorem.

Allocation and source. This is the early sector checkpoint in planned RRO1. Full contract: QEO1/T06. QEM1 Sections 19–20. Count its proof once in the 11–15-page T05–T07 budget; TPO1 imports the identity.

14. T07. The exact source-shift and Wick identity

Model-specific. Overview Target.

Required result. For the finite Gaussian and determinant representation identified by T06, shift the complex external source and derive the exact Wick derivative identity. Retain the normalizing denominator, source variance and powers of the physical charge.

Application inputs. Use T05 for the actual orthogonal reflected probes. Distinguish the general finite Gaussian identity from these model-specific source checks. The result converts the normalized determinant derivatives in Q4R into the reference Wick products used in T17.

Allocation and source. This completes the early identity part of planned RRO1. Full contract: QEO1/T07. QEM1 Section 20. Its proof belongs to the combined 11–15-page source/sector/Wick assignment.

15. Q4R. Normalized reference response

Model-specific. Overview Target.

Required result. For the reference torus and the particular bare pair selected by T01, bound the normalized mixed derivatives after subtracting the free one-cell coefficients. Keep the entire finite set of reflected cells and the common normalization. The estimate is uniform in fine mesh and permitted placement.

Main proof work. Using the terminal analytic and measure outputs in T09, prove the one-cell determinant estimate T12, the marked multi-cell bounds T13, and the positive small-field representation with controlled complex complement T14. Only then apply the finite-jet cancellation T15. The source identity is T07. The pressure estimate T11 is a separate input to the later volume transfer.

Allocation and contracts. This is the later checkpoint in planned RRO1. Full statement: QEO1/Q4R; proof assignments T12– T15. QEM1 Section 23, pp. 227–243. Allow 16–22 pages for this response argument in addition to the early identity assignment.

16. Q1. Finite all-flux inputs for volume transfer

Model-specific. Overview Target.

Required imports. Use the chamber finite-volume positivity and Gaussian domination, rectangular chessboard inequalities and the trace comparison specified by QEO1/Q1. All are applied to the same charge and total bare mass as the reference construction.

Boundary of this input. The infinite-lattice RP conclusion by itself does not supply every finite-volume inequality used in the transfer. The finite Q1 estimates remain explicit imports. The quantitative photon determinant defect and flux-tail comparisons are additional proof tasks inside T17.

Source and related work. QEO1/Q1 and QEO1/T17; QEM1 Sections 2–5 and 25. LLO1 and LPO1 organize the related fixed-spacing work under IVO1. They remain outside this umbrella; their results enter at the indicated application.

17. T17. Uniform products on arbitrary construction volumes

Model-specific. Overview Target.

The workhorse estimate. Let $F_\eta^\sharp$ denote the physical cutoff field, f^η the source test and v_η its variance. For $m = 2, 4$, put

$$H_2(x; v) = x^2 - v, \qquad H_4(x; v) = x^4 - 6vx^2 + 3v^2.$$

For every finite set Y of distinct translates on the specified probe lattice, prove

$$\left| \mathbb{E}^{\mathrm{af}}_{N,T} \prod_{y \in Y} [e^{-m} H_m(F_\eta^\sharp(\tau_y f^\eta); v_\eta) - c_{m,\eta}] \right| \le u_{m,\eta}^{|Y|},$$

where e^{-m} is an inverse power of the physical charge, and

$$u_{4,\eta} \le Ce^\sigma, \qquad (1 + |c_{2,\eta}|)u_{2,\eta} \le Ce^\sigma.$$

The constants are independent of construction volume and fine mesh above the stated thresholds. The exact lattice, volume arithmetic, thresholds and charge-exponent loss are part of QEO1/T17.

Inputs that must meet. Combine the reference response Q4R, common two-volume pressure T11, free coefficient comparison T04, and finite positivity and chessboard estimates Q1. Prove the photon determinant defect and flux-tail comparisons inside this transfer argument. Every interacting reference and outer-volume law must use the pair selected by T01; independent admissible choices would not justify the cancellation. The free coefficient comparison retains its unshifted positive reference mass.

Allocation. This is the first checkpoint in planned VTO1. The second is the same-component passage A3, using B5. QEM1 Section 25, pp. 247–255: allow 9–13 pages here. The combined response statement is QEO1/Q4.

18. A1. Continuum parents, recovery and moment limits

Model-specific. Overview Target.

External input. For each prescribed volume-first parent, retain its identity through a joint cutoff realization. Use uniform Gaussian domination to pass every fixed translated product, replace the cutoff tests by their common L^2 limit, and retain the needed joint moments and cumulants.

Current treatment and application check. Current CEM1 supplies continuum compactness, Gaussian and correlation convergence, structural inheritance and prescribed-volume recovery. The response application must also verify moment convergence along its finite-torus diagonal using the finite periodized Gaussian estimate. That check belongs to the T18 application, not to a new compactness theorem.

Use and contracts. Apply these statements to the same bare sequence as T17. Their use in A3 preserves the already chosen parent; a newly selected unrelated limit would not suffice. Full master contracts: QEO1/A1, B3, and T18. The structural branch remains organized directly under QEO1.

19. B2. Gaussian domination in the same components

Interface. Overview Target.

External theorem. For a translation-invariant continuum law with invariant coordinate and diagonal reflection forms, pass those forms to almost every translation-ergodic component on one common full-measure set. If the parent satisfies an L^2 Gaussian bound, the components retain the same constant.

Use here. The component Gaussian bound gives centering and the moments needed in the conditional-cumulant calculation. Apply it to the same parent realized in A1. The theorem also supplies clustering for the structural OS branch.

Existing treatment. ECM1 is the focused continuum-component manuscript. The full contract is QEO1/B2. This is an external input to RGO1; no additional lattice ergodicity project is introduced.

20. B5. Distinct products control conditional expectations

Interface. Overview Target.

General input. For a measure-preserving lattice action and a random variable with all finite moments, uniform bounds on every product at distinct translates control its invariant conditional expectation. Apply the result to each centered Wick probe. No ergodicity of this discrete lattice action is assumed.
Application. After A1 passes the finite products to the fixed continuum parent, apply the criterion at Wick orders two and four. The all-translation invariant sigma-algebra is contained in the probe-lattice invariant sigma-algebra. Use B2 for conditional centering, then recover the fourth cumulant from the two conditional Wick moments.
Allocation and contracts. This interface is proved once inside planned VTO1. QEO1/B5 gives the criterion; M19 its invariant-product proof; T18 the application. QEM1 Section 26, pp. 255–258. Allow 4–7 pages for the whole conditional-cumulant assignment.

21. T16. One common choice of constants and charge range

Model-specific. Overview Target.

Required verification. Choose the blocking, localization and small-field constants in the order specified by QEO1/T16. After fixing the constants in the conditional kernel, nonlinear and response estimates, verify that their inequalities hold on one positive charge interval. Retain the support-placement and mesh thresholds.

Meaning of the arrows. The incoming arrows carry the rates and numerical conditions in the conditional estimates. This verification permits their simultaneous use in the final conclusion. It is not an input to the proofs of those conditional estimates and creates no cycle in the dependency graph. The final probe-dependent restriction for positivity is made after the uniform cumulant error constant is fixed.

Allocation and source. This shared verification remains within the umbrella and the consuming model proofs; no separate focused work is allocated. Full contract: QEO1/T16. QEM1 Section 24, pp. 244–247.